StreamFlix Final Client Report

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1. Executive Summary

StreamFlix engaged Team 1 to analyse customer, movie and ratings data to support its transition from a free ad-supported platform to a subscription-based streaming service. We designed an end-to-end analytics solution using SQL Server, Python and Power BI to deliver interactive reporting and business recommendations.

2. Business Problem

The client required insights into:
• Top-performing movies and genres
• User preferences by age, country and subscription type
• Customer engagement patterns
• Interactive reporting for decision-making

3. Project Objectives

• Clean and validate the datasets
• Build a relational SQL database
• Perform analytical SQL queries
• Generate Python visualisations
• Develop an interactive Power BI dashboard
• Deliver actionable business recommendations

4. Project Methodology

Workflow:
Client CSV Files
↓
ETL (Extract, Transform, Load)
↓
SQL Server Database
↓
SQL Analysis
↓
Python Visualisations
↓
Interactive Power BI Dashboard
↓
Business Insights & Recommendations

5. Dataset Overview

- **Movies**: Movie information including title, genre, language, country and total views.
- **Users**: Customer demographics, subscription status, device and watch time.
- **Ratings**: User ratings linked to movies through primary and foreign keys.

6. ETL Process

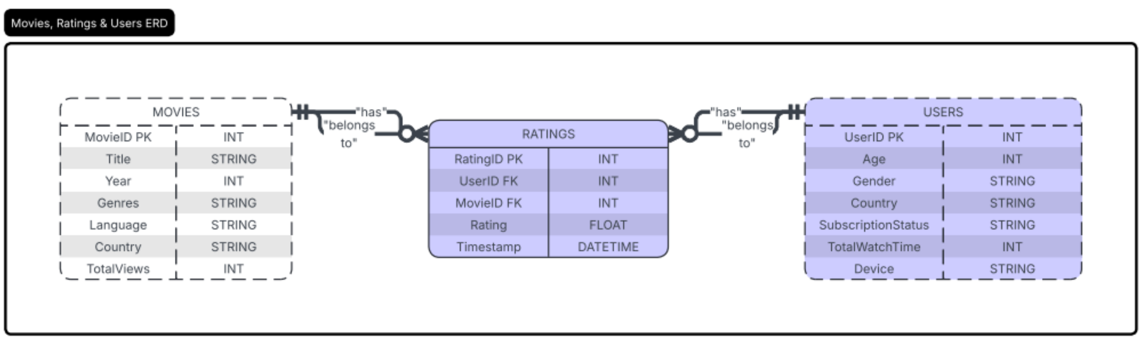
- **Extract**: Imported Movies.csv, Users.csv and Ratings.csv.
- **Transform**: Removed duplicates, checked missing values, standardised formats, validated data types, extracted movie release year and verified relationships.
- **Load**: Imported cleaned data into SQL Server and established primary and foreign keys.

7. SQL Database Design

Created three related tables:

- Movies
- Users
- Ratings

Relationships:
Users (1) → Ratings (Many)
Movies (1) → Ratings (Many)
Ratings links users with movies.



8. SQL Analysis

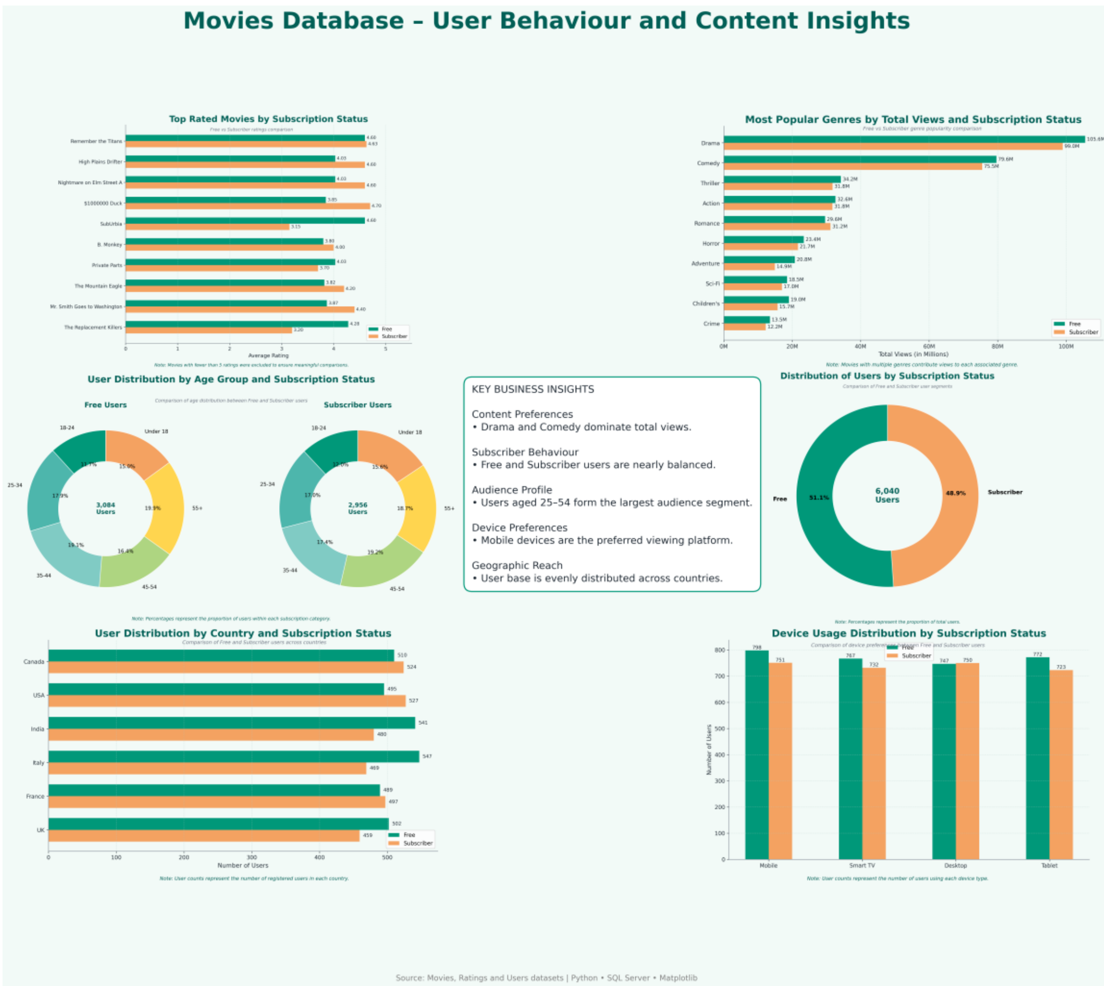
Analytical queries answered business questions including:

- Average ratings by subscription type
- Top-rated movies
- Most popular genres
- User engagement by country
- Device usage
- Viewing patterns

9. Python Visualisations

Python (Pandas and Matplotlib) was connected to SQL Server to generate:

- Top 10 Movies by Average Ratings
- Most Popular Genres
- User Distribution by Age Group
- Distribution of Users by Subscription Status
- User Distribution by Country
- Distribution of Device Usage



10. Power BI Dashboard

The interactive dashboard contains:

- Presentation Roadmap
- Python Visualisations
- Interactive Business Insights Dashboard
- Global Users Market and Behavioral Nuance
- Subscription status and Age Distribution.
- Movie Insights
- Business Recommendations

Interactive slicers:

- Subscription Status
- Country
- Age Group

11. Key Business Insights

Based on our analysis of the Movies, Users, and Ratings datasets, we identified several insights that can help StreamFlix improve customer engagement, increase subscriber growth, and support its transition to a subscription-based platform.

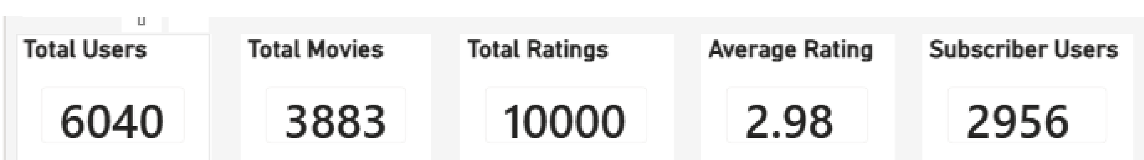
11.1 Content Portfolio Performance

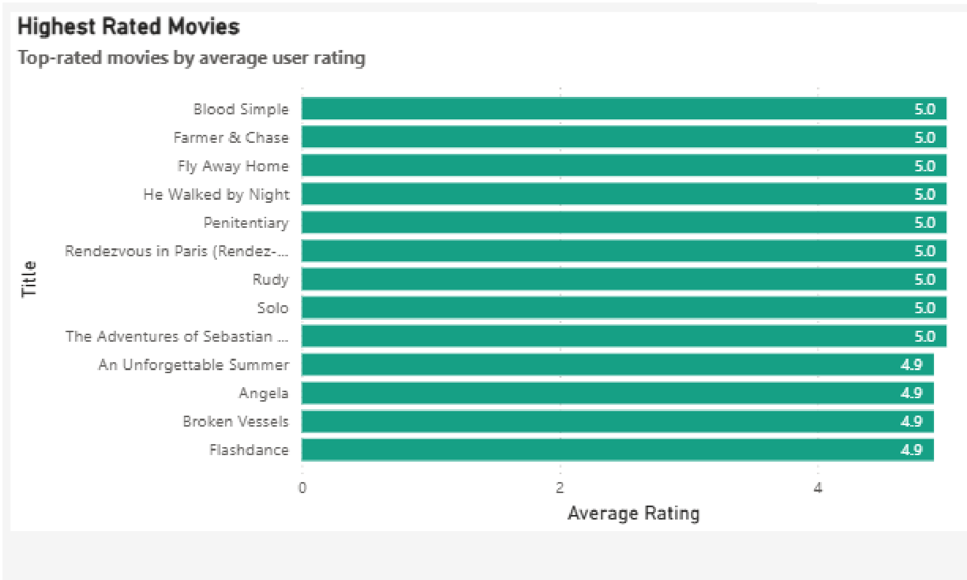
Key Insights

- The highest-rated movies consistently received strong customer ratings, indicating content that should be retained and promoted.
- Drama, Comedy, and Action emerged as the most popular genres based on audience engagement.
- Investing in consistently high-performing genres can maximise customer satisfaction and viewing time.

Business Value

Focusing on proven content helps optimise licensing investments while improving customer retention and platform engagement.





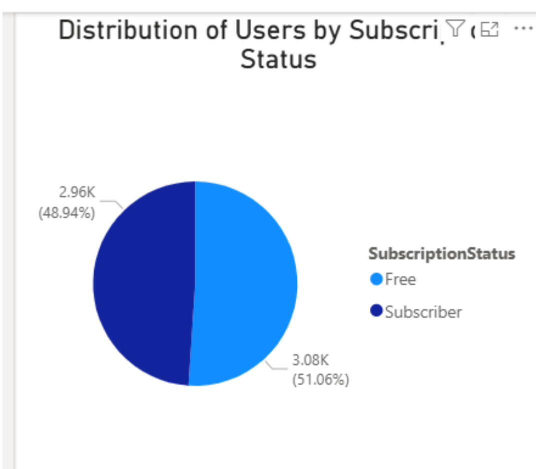
11.2 Audience Composition

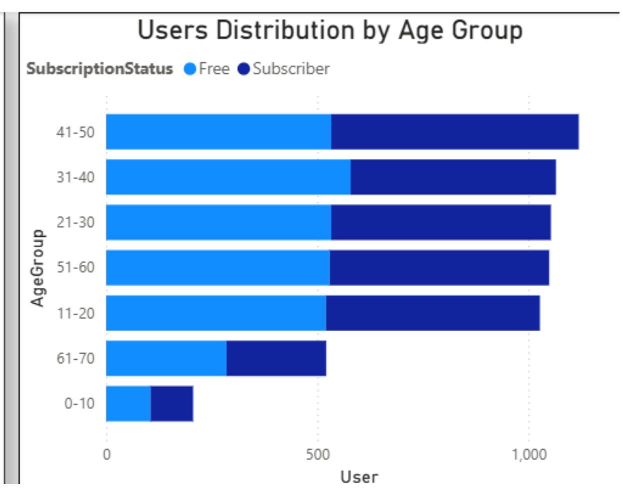
Key Insights

- Users are distributed across multiple age groups, with the strongest representation in the 21–60 age range.
- Subscription behaviour is relatively balanced across age groups, indicating opportunities to personalise experiences for different customer segments.

Business Value

Understanding audience demographics enables StreamFlix to create targeted recommendations and age-specific marketing campaigns.





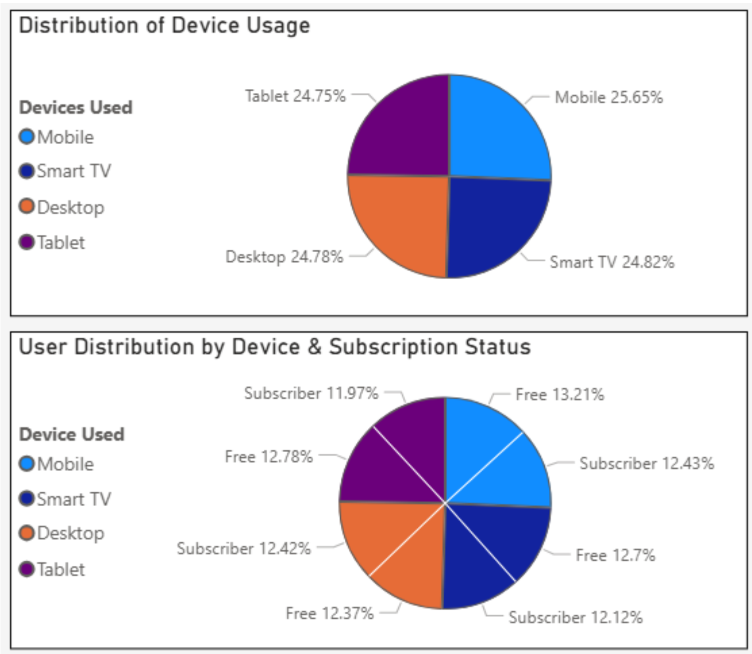
11.3 Consumption Behaviour

Key Insights

- The platform currently maintains a balanced mix of Free and Subscriber users.
- Device usage is distributed across Mobile, Desktop, Smart TV, and Tablet platforms, with Mobile representing a significant access channel.

Business Value

Improving the mobile experience and encouraging free users to upgrade to premium subscriptions can increase engagement and revenue.



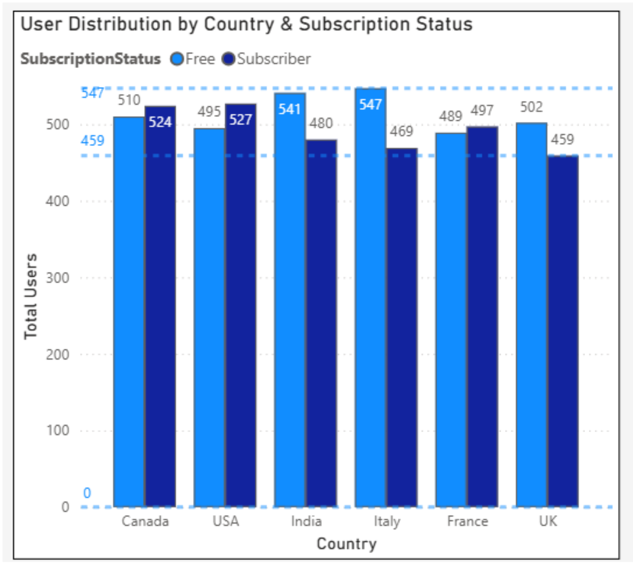
11.4 Geographic Reach

Key Insights

- StreamFlix has an established audience across multiple countries.
- User engagement varies by region, highlighting opportunities for market-specific content and promotional strategies.

Business Value

Country-level insights support localisation strategies, regional marketing campaigns, and international growth.



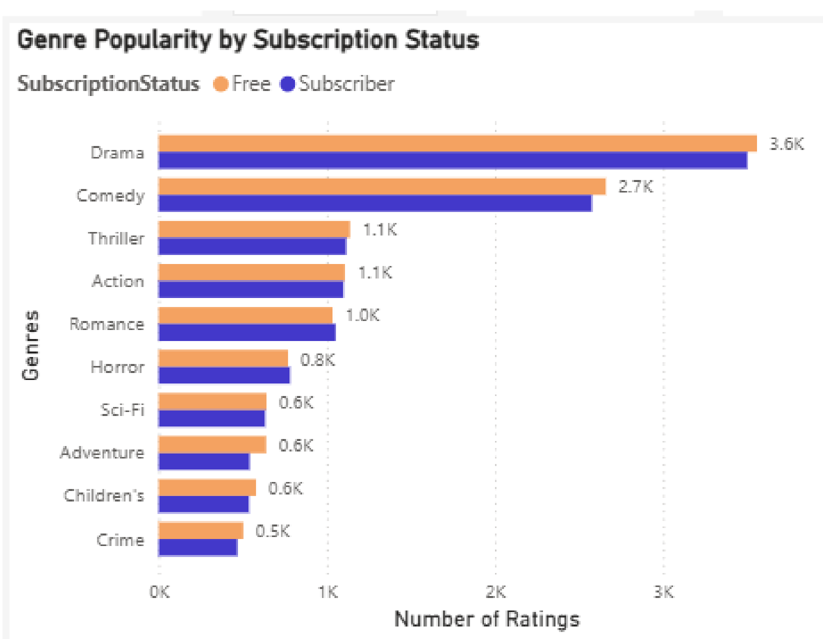
11.5 Performance Drivers

Key Insights

- Average audience ratings indicate opportunities to improve overall content quality.
- Drama, Comedy, and Action continue to be the strongest-performing genres.

Business Value

These insights guide future content acquisition and licensing decisions while improving recommendation accuracy.



11.6 Executive Priorities

The analysis highlights four strategic priorities:

- Understand customer behaviour.
- Invest in content that consistently performs well.
- Increase conversion from Free users to Subscribers.
- Optimise the viewing experience across all devices.

12. Strategic Recommendations

Based on our findings, Team 1 recommends the following actions to support StreamFlix's business objectives.

Recommendation 1 – Invest in High-Performing Content

Continue investing in genres that consistently receive high ratings and strong engagement, particularly Drama, Comedy, and Action.

Expected Benefit

- Higher customer satisfaction
- Better return on content investment
- Increased viewing hours

Recommendation 2 – Retain and Promote Top-Rated Movies

Protect licensing agreements for highly rated titles and feature them prominently within the platform.

Suggested Actions

- Homepage recommendations
- Featured collections
- Personalised movie suggestions

Expected Benefit

Improved customer retention and stronger platform engagement.

Recommendation 3 – Personalise the Customer Experience

Use demographic insights such as age, country, and subscription status to deliver tailored content recommendations.

Expected Benefit

- Higher watch time
- Improved recommendation relevance
- Better customer satisfaction

Recommendation 4 – Increase Subscriber Conversion

Develop initiatives that encourage Free users to become paying subscribers.

Suggested Strategies

- 30-day premium trial
- Exclusive subscriber-only content
- Personalised upgrade offers
- Early access to new releases

Expected Benefit

Higher subscription revenue and improved customer retention.

Recommendation 5 – Expand in High-Growth Markets

Focus investment on countries with strong engagement and growth potential.

Suggested Strategies

- Localised content
- Regional marketing campaigns
- Multi-language subtitles
- Country-specific recommendations

Expected Benefit

Support global expansion while increasing customer acquisition.

Recommendation 6 – Optimise Platform Performance

Prioritise improvements for the most frequently used devices.

Focus Areas

- Mobile application performance
- Smart TV optimisation
- Cross-device continuity
- Streaming quality improvements

Expected Benefit

A better user experience that increases engagement and long-term loyalty.

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13. Future Opportunities

While this project addressed the current business requirements, additional analytics could further strengthen StreamFlix's decision-making capabilities.

Future enhancements include:

- Machine Learning recommendation engine
- Customer churn prediction
- Personalised content ranking
- Sentiment analysis using customer reviews
- Real-time dashboards connected to live streaming data
- Predictive analytics for future content demand

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14. Conclusion

This project demonstrates how an end-to-end analytics solution can transform raw data into meaningful business insights.

Using SQL Server, Python, and Power BI, Team 1 developed an interactive reporting solution that enables StreamFlix to better understand customer behaviour, evaluate content performance, and support strategic decision-making.

By implementing the recommendations outlined in this report, StreamFlix can:

- Improve customer engagement
- Increase subscriber conversion
- Strengthen customer retention
- Optimise content investment
- Expand into high-growth markets
- Deliver a better streaming experience across all devices

The combination of robust data preparation, analytical reporting, and interactive dashboards provides a scalable foundation for future growth and continued data-driven decision-making.

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15. Technologies Used

SQL Server, SQL, Python (Pandas, Matplotlib), Jupyter Notebook, Microsoft Visual Studio, Microsoft Power BI.
